

Lecture 10 Tutorial — AI Applications

Q1. AI System Pipeline (Ordering)

Arrange the following stages into the correct order of an AI system workflow:

- A. Feature Extraction
- B. System Deployed
- C. Collect Data
- D. Machine Learning
- E. Formulate Problem

Write your answer as an ordered sequence (e.g., $E \rightarrow C \rightarrow \dots$).

Q2. Intelligence Category (Matching)

Match each example to the most appropriate category.

Categories:

- A. Computation Intelligence (Feature $\rightarrow$ Label, unified)
- B. Perception Intelligence (Feature $\rightarrow$ Labels, non-unified)
- C. Creative Intelligence (beyond current generative models)

Examples:

1. Outlier detection
2. Emotion detection
3. Creative writing / story generation

Write: 1-_____, 2-_____, 3-_____.

Q3. Computer Vision Tasks vs Applications (Matching)

Match each application to the most essential computer vision task.

Tasks:

- A. Object detection/recognition
- B. Image segmentation
- C. Motion analysis
- D. Scene understanding (Q&A system)

Applications:

1. Autonomous vehicles
2. Medical imaging (tumour boundary delineation)
3. Surveillance systems (tracking movement in a scene)
4. Visual question answering over images

Write: 1-_____, 2-_____, 3-_____, 4-_____.

Q4. Traditional vs Deep Learning in Vision (Multiple Choice)

Choose the most appropriate method family for each statement.

1. “Designed as a classic feature-based detector for faces in real time.”
 - A. Canny/Sobel edge detection
 - B. Haar cascades
 - C. Vision Transformer (ViT)
 - D. YOLO
2. “End-to-end model class that learns convolutional filters for images.”
 - A. CNN
 - B. Sobel
 - C. Haar cascades
 - D. TF-IDF
3. “Modern architecture using attention mechanisms for vision.”
 - A. CNN
 - B. ViT
 - C. HMM
 - D. NMF

Q5. Tool Selection (Multiple Choice)

Choose the best tool/library for each task (single best answer).

1. Classical computer vision prototyping (image I/O, filtering, detection):
 - A. OpenCV
 - B. Kaldi
 - C. NLTK
 - D. Librosa
2. Object detection training/evaluation framework in deep learning:
 - A. MMDetection
 - B. Audacity
 - C. Stanford CoreNLP
 - D. TF-IDF
3. Speech recognition toolkit (ASR pipeline):
 - A. Kaldi
 - B. OpenCV
 - C. MMDetection
 - D. Spacy

Q6. Audio Tasks and Methods (Matching)

Match each task to a method that is explicitly associated with it in the lecture content.

Methods:

- A. MFCCs
- B. HMMs
- C. NMF
- D. RNN/LSTM
- E. Transformer model (e.g., Whisper)
- F. Diffusion models

Tasks:

- 1. Speech feature extraction
- 2. Sequential audio modelling (classic approach)
- 3. Source separation
- 4. Temporal dependencies for speech-to-text
- 5. Multilingual ASR (modern approach)
- 6. High-fidelity music synthesis

Write: 1-_____, 2-_____, 3-_____, 4-_____, 5-_____, 6-_____.

Q7. Audio Application Mapping (Multiple Choice)

Pick the most appropriate key task for each application.

- 1. Voice biometrics for access control:
 - A. Speaker identification/verification
 - B. Beat/tempo detection
 - C. Noise cancellation
 - D. Image segmentation
- 2. Gunshot detection in surveillance:
 - A. Environmental sound detection
 - B. Machine translation
 - C. Object recognition
 - D. Creative writing
- 3. Hearing aids with AI-driven sound enhancement:
 - A. Noise cancellation / sound enhancement
 - B. Dependency parsing
 - C. Scene understanding
 - D. Outlier detection

Q8. NLP Levels (Classification)

Classify each task as Syntax-level or Semantic-level.

1. Tokenization / POS tagging
2. Dependency parsing
3. Named Entity Recognition (NER)
4. Sentiment analysis
5. Machine translation

Write your answers as: 1-_____, 2-_____, 3-_____, 4-_____, 5-_____.

Q9. Traditional vs Deep Learning in NLP (Multiple Choice)

For each item, choose whether it is Traditional or Deep Learning (single best answer).

1. Bag-of-Words (BoW)
 - A. Traditional
 - B. Deep Learning
2. TF-IDF weighting
 - A. Traditional
 - B. Deep Learning
3. Transformer models (BERT, GPT)
 - A. Traditional
 - B. Deep Learning
4. RNNs/LSTMs
 - A. Traditional
 - B. Deep Learning

Q10. Multi-modal Processing Pipeline (Fill in the blanks)

Complete the sequence using these components:

{Multi-modal Input, Feature Extraction, Text Features, Audio Features, Visual Features, Fusion Layer, Decision Output}

Write the correct pipeline:

_____ → _____ → (_____, _____, _____) → _____ → _____

Q11. Multi-modal Core Challenges (Select all that apply)

Which items are listed as core challenges for multi-modal media processing? Select all that apply.

- A. Cross-modal alignment (e.g., audio-visual synchronization)
- B. Feature fusion strategies
- C. Context-aware interpretation
- D. Chi-square test for variance
- E. Laplacian correction

Q12. Application-to-Subfield-and-Task (Structured Selection)

For each scenario, choose (i) the most relevant AI subfield and (ii) one key task.

Subfields (choose one each):

- A. Computer Vision
- B. Computer Auditory Systems
- C. NLP
- D. Multi-modal Media Processing

Tasks (choose one each):

- Object detection/recognition
- Automatic Speech Recognition (ASR)
- Text summarization
- Cross-modal alignment / feature fusion

Scenarios:

1. “Automated lecture transcription with slide analysis.”
2. “Real-time background noise suppression in video calls.”
3. “A chatbot that answers questions and summarizes long documents.”
4. “Traffic monitoring that detects vehicles in live video.”

Answer format (example):

Subfield __, Task __

1. Subfield _____, Task _____
2. Subfield _____, Task _____
3. Subfield _____, Task _____
4. Subfield _____, Task _____